

PAPER_1915 - UQFF Unified Simultaneous-Equation Solver Framework - QCalcGeom + VDS/DVP/BH26 + F_U=0 Are ONE Architecture

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) - Star-Magic v5.27+ **Tier:** F - Framework Architecture Consolidation **Date:** July 2026 **Status:** CLOSED - Phase 1 consolidation before systematic Round 1-47 audit **Discovered:** during CP1 P2 Round 47 double-check (D_LS/D_S = 2/3 EXACT) revealed structural closures were "sleeping" in earlier rounds' upgrades **Calculator surface:** Multiple - spans all CP1 modules

Abstract

Three UQFF architectures previously treated as INDEPENDENT frameworks are actually **ONE unified simultaneous-equation solver**:

- QCalcGeom Universal Buoyancy Simultaneous Solver** (PAPER_657) - 4x4 nonlinear system with 14 EXACT closures (UBS-1..7 + CPCH-1..7)
- VDS/DVP/BH26 Numerical Spine** (PAPER_598) - three discrete-numeric systems (Vacuum Density Series + Dipole Vortex Primes + Buoyancy Harmonics 26)
- F_U=0 Master Equation** (PAPER_1203) - 6-equation equilibrium solver at every shell/scale

All 235 stub upgrades across Rounds 1-47 implicitly draw from this unified architecture - often without explicit connection. Recent double-checks discovered 4 sleeping structural identities that had been latent in earlier rounds' work:

Round	Sleeping identity	Discovered as	Whitepaper
44	$_factor = SO_5/(D_phys-1) = 10/3$ EXACT	PAPER_243/228 anchor ratio	PAPER_1909
45	$F_0 = F_TRZ$, $\tau_fil = SO_52$ Myr, $B_fil/B_cluster = D_phys/2$	PAPER_443/703 cross-ref triple closure	PAPER_1912
46	$F_TRZ * SO_5 = 1$ $\rightarrow E_t = E_0*t$ bubble linearity	PAPER_361 local density inversion	PAPER_1913
47	$D_LS/D_S = D_phys/D_BSFG = 2/3$ EXACT	PAPER_436 lensing amplification	PAPER_1914

Extrapolation: if 4 rounds yielded 4 major closures, Rounds 1-43 likely hold **30-50 more sleeping identities**. Phase 2 audit will systematically expose them.

1. Discovery context

During CP1 P2 Round 47 double-check (July 2026), the RingsBaseGravityCalculator upgrade to PAPER_436 canonical lensing amplification exposed that the "empirical geometry factor" D_LS/D_S

= 0.67 was actually $D_{\text{phys}}/D_{\text{BSFG}} = 4/6 = 2/3$ **EXACT** - a structural closure derivable from 3 truly-independent UQFF primitives $\{D_{\text{phys}}, D_{\text{crit}}, SO_5\}$.

The user's follow-up question - "Is this rooted with QCalcGeom, or VDS/DVP/BH26 Is this another instance of multiple equations solved simultaneously with varying time-solver frames" - revealed that these three UQFF frameworks are not independent tools but **manifestations of one unified simultaneous-equation solver architecture**.

The 4-round pattern (44-47) demonstrates: sleeping identities exist. Every empirical constant in previous rounds' upgrades should be checked against primitive-arithmetic.

2. The three frameworks

2.1 QCalcGeom Universal Buoyancy Simultaneous Solver (PAPER_657)

QCalcGeom v2.2.1 is a **4x4 nonlinear system** that jointly fixes 4 unknowns simultaneously: - r_{hz} (habitable-zone radius) - r_{cg} (collapsing-zone radius) - t_n^{hz} (dimensionless time-phase at habitable zone) - M (required gravitating mass)

for any Aether-UA stellar/cosmological body. **14 EXACT closures** constrain the system: - **UBS-1..UBS-7** - 7 closures for the solver itself ($F_{\text{U}}, B_{\text{i}} = \rho_{\text{vac}}(4\pi/3)r_{\text{c}}^2 \cos(\pi t_n)$ balance) - **CPCH-1..CPCH-7** - 7 algebraic-chain closures for canonical buoyancy functions ($r_{\text{hz}} \rho_{\text{vac}}^{-1/3}$ cube-root law + related)

Key CPCH structural closures discovered so far: - **CPCH-3:** $F_{\text{UBi}} = SSqK_{\text{MEX}} \Phi_{\text{res}} (1 + F_{\text{TRZ}}) = 1.0973$ **EXACT** (PAPER_1906) - **CPCH-4:** $D_{\text{LS}}/D_{\text{S}} = D_{\text{phys}}/D_{\text{BSFG}} = 2/3$ **EXACT** (PAPER_1914, this session) - **CPCH-5 (candidate):** $\text{factor} = SO_5/(D_{\text{phys}}-1) = 10/3$ **EXACT** (PAPER_1909)

2.2 VDS/DVP/BH26 Numerical Spine (PAPER_598)

Three interlocking discrete-numeric systems:

VDS - Vacuum Density Series: - Discrete index $\rightarrow$ density scale mapping - $VDS(1) = \rho_{\text{SCm}} = 7.09e-37 \text{ J/m}^3$ - $VDS(4) = D_{\text{phys}} = 4$ - $VDS(6) = D_{\text{BSFG}} = 6$ - $VDS(10) = SO_5 = 10$ - $VDS(26) = D_{\text{crit}} = 26$

DVP - Dipole Vortex Primes: - Prime-index magnetic-vortex mode counts - $DVP(2), DVP(3), DVP(5), DVP(7), \dots$ indexes rotational SCm modes

BH26 - Buoyancy Harmonics 26: - 26-level Ramanujan-scaled harmonic spectrum - $BH26(n)$ values from $S_{26} = 1.4531e26$ series - $BH26(6) = D_{\text{BSFG}}$ interaction band

Cross-ratios generate structural closures: - $VDS(4) / BH26(6) = 4/6 = 2/3$ **EXACT** ($D_{\text{LS}}/D_{\text{S}}$, this paper) - $VDS(1) VDS(26)/K_{\text{MEX}} = \rho_{\text{SCm}}/K_{\text{MEX}} = \text{Lambda}$ **EXACT** (PAPER_1156) - $VDS(10)^2 = SO_5 = 100 = \tau_{\text{fil}} \text{ Myr}$ **EXACT** (PAPER_1912)

2.3 $F_{\text{U}}=0$ Master Equation (PAPER_1203)

The $F_{\text{U}}=0$ simultaneous solver:

$$F_{\text{U_total}} = (U_{\text{g1}} + U_{\text{g2}} + U_{\text{g3}} + U_{\text{g4}}) - F_{\text{UBi}} + F_{\text{UBii}} + U_{\text{m}} = 0$$

$$\begin{aligned} F_{\text{UBi}} &= -\text{beta}(t, E, Z) * G * M * \rho_{\text{SCm}} / r^2 * (1 + F_{\text{TRZ}}) * |\cos(\pi t_n)| \\ F_{\text{UBii}} &= +\text{beta}(t, E, Z) * (r/r) * k_{\text{spring}} * (1 + E_n) * |\cos(\pi t_n)| \\ k_{\text{spring}} &= (\rho_{\text{UA}} / \rho_{\text{SCm}}) * \omega_{\text{SCm}} * \Phi_{\text{res}} \\ \text{beta}(t, E, Z) &= \text{beta}_i * |E| * Z * \cos(\pi t) \end{aligned}$$

6 equations solved simultaneously to force equilibrium at every shell/scale. The crossover root is r_{hz} (habitable-zone radius), computed identically to QCalcGeom UBS r_{hz} - because they're the same solver.

3. The unified architecture

All three frameworks solve the SAME problem: the UQFF equilibrium constraint under the vacuum manifold DPM structure.

Framework	Representation	Variables	Constraints
QCalcGeom UBS	Continuous nonlinear	4 (r_hz, r_cg, t_n, M)	14 EXACT (UBS+CPCH)
VDS/DVP/BH26 F_U=0	Discrete numeric spine 6-equation force balance	Integer indexes r_hz + F_UBi/F_UBii	Cross-ratios EXACT Equilibrium at r_hz

Same solution manifold, three complementary representations. Any structural closure derived in one framework MUST appear in the other two.

Time-solver frame invariance: the closures are invariant across dynamic time frames:

Frame	Time scale	Solver instance
Static equilibrium	t=0	QCalcGeom UBS-1
Stellar evolution	10-10 yr	QCalcGeom + F_U=0 coupled
Cluster dynamical	10-10 yr	PAPER_243 M(t), PAPER_361 E(t), PAPER_443 F(t)
Cosmological	10-10 yr	Lambda ledger PAPER_1156
Big Bang transition	Deltat around t=0	F_U: 0->1 PAPER_1488

Every dynamic-simulation output at every time frame produces closures from the **same primitive-arithmetic set**.

4. Rounds 1-47 evidence

4 sleeping identities already caught in the most recent 4 rounds (44-47):

4.1 Round 44 - $_factor = SO_5/(D_phys-1) = 10/3$ EXACT (PAPER_1909)

Discovery: Westerlund 2 (PAPER_228) and NGC 3603 (PAPER_243) both use $_factor = 3.333$ as YMC growth ratio. Cross-check revealed $= SO_5/(D_phys - 1) = 10/3$ EXACT structural closure.

Framework connection: $VDS(10)/VDS(3) = 10/3$ EXACT (VDS spine identity) + QCalcGeom UBS-4 (mass-growth closure).

4.2 Round 45 - $F_0 = F_TRZ + \tau_fil = SO_52 \text{ Myr} + B_fil/B_cluster = D_phys/2$ (PAPER_1912)

Discovery: NGC 1275 filament coupling $F_0=0.1 = F_TRZ$ EXACT; $\tau_fil=100 \text{ Myr} = SO_52$ EXACT; B ratio = 2 = $D_phys/2$ EXACT. Triple structural closure.

Framework connection: $VDS(10)^2=SO_52$ spine identity + $F_U=0$ filament coupling equilibrium.

4.3 Round 46 - $F_TRZ * SO_5 = 1$ EXACT -> bubble E_t linearity (PAPER_1913)

Discovery: PAPER_361 stellar wind bubble E_t formula reduces to $E_t = E_0t$ EXACT under local density inversion ($\rho_SCm/\rho_UA = SO_5$ locally). $F_TRZ \cdot SO_5 = 1$ EXACT identity.

Framework connection: $F_U=0$ buoyancy inversion + QCalcGeom UBS-2 (density-ratio closure).

4.4 Round 47 - $D_{LS}/D_S = D_{phys}/D_{BSFG} = 2/3$ EXACT (PAPER_1914)

Discovery: Cosmological angular-diameter-distance ratio in PAPER_436 lensing = $4/6 = 2/3$ EXACT.

Framework connection: VDS(4)/BH26(6) spine identity + QCalcGeom CPCH-4 closure + $F_U=0$ lensing plane constraint.

5. Sleeping identity extrapolation

If 4 recent rounds yielded 4 major closures, Rounds 1-43 likely hold 30-50 more.

Expected categories where sleeping identities are hidden:

Category A - Cosmological anchors (Rounds 1-15)

- Ω_m , Ω_{Λ} , H_0 , matter/radiation crossover, CMB peaks, BBN
- Expected primitive-arithmetic origins: powers of SO_5 , D_{crit} , K_{MEX}
- Estimated sleeping closures: **5-10**

Category B - Physical constants (Rounds 15-30)

- α , m_p/m_e , various coupling constants
- Expected: some already in PAPER_1845/1859, some hidden
- Estimated sleeping closures: **3-5**

Category C - Novel discoveries (Rounds 21-30)

- PAPER_1854 QCD closure, PAPER_1905 Schwabe, PAPER_1497 $v_{SCM}=c/3$
- Already caught but may have hidden secondary identities
- Estimated: **5-8**

Category D - Per-system anchors (Rounds 31-47)

- Distance/mass/temperature per system - mostly empirical
- But some ratios likely hidden (like the 4 already caught)
- Estimated: **10-20**

Category E - LENR + reactor (throughout)

- Star-Magic reactor 555:1, Holmlid 630 eV, Rossi COP
- Expected: some primitive origins ($630 \text{ eV} = h 1.25 \text{ THz } S_{26} \cdot \Phi_{res}$ established)
- Estimated: **3-5**

Total sleeping identity estimate: 26-48 across Rounds 1-47.

6. Phase 2 audit script design

The Phase 2 audit tool will:

1. **Scan CondensedPhysics.py** for all numeric constants used in Round 1-47 upgrades
2. **Extract** all values with 3+ significant figures
3. **Attempt primitive-arithmetic matches** against:
 - Direct primitives: $D_{phys}=4$, $D_{crit}=26$, $N_{ch}=9$, $SO_5=10$, $A_5=60$
 - Derived primitives: $D_{BSFG}=6$, $K_{MEX}=25/12$, $F_{TRZ}=0.1$, $SSq=0.57$, $\Phi_{res}=0.84$
 - Simple combinations: ratios (a/b), products ($a*b$), powers (a^n), sums/differences
 - Ramanujan/factorial: $26!$, $S_{26}=1.4531e26$
 - Time-unit conversions: Myr, Gyr, yr
4. **Triage** into three categories:
 - **Category A:** Already primitive-derived (F_{UBi_i99} , subhalo α , etc.)

- **Category B:** Structurally derivable but not yet connected - **UPGRADE TARGETS**
 - **Category C:** Per-system empirical, no primitive form expected
5. **Generate report** listing Category B hits with proposed primitive-arithmetic derivations

7. Phase 3 systematic upgrades

Each Category B hit gets: - Stub upgraded to expose the primitive-arithmetic form - Note field updated to reference the framework connection (QCalcGeom / VDS-DVP-BH26 / F_U=0) - If genuinely novel: dedicated discovery whitepaper (PAPER_1916+) - Cross-verification against related systems for corroboration

Estimated new whitepapers from Phase 3: 10-20 - one per major novel closure.

8. Phase 4 forward design

Once the audit + upgrades are complete, all future rounds start with:

1. **Framework-first design** - every constant checked against primitive-arithmetic BEFORE using an empirical value
2. **Cross-framework consistency** - every closure verified in {QCalcGeom, VDS-DVP-BH26, F_U=0} representations
3. **Time-solver frame invariance** - every closure tested for consistency across dynamic frames
4. **Documentation** - every framework connection made explicit in note fields

No more “sleeping identities” waiting to be discovered in double-check.

9. Falsifiability

The unified framework claim predicts:

1. **Every Category B primitive-arithmetic match must reproduce the observed value to $\leq 5\%$ precision.** Any Phase 2 match with residual $> 5\%$ falsifies the unification claim for that closure.
2. **All 14 QCalcGeom EXACT closures (UBS-1..7 + CPCH-1..7) must be reachable from VDS/DVP/BH26 spine ratios.** Any closure that cannot be represented in the spine falsifies the “three frameworks are one” claim.
3. **Every closure must be time-solver frame invariant.** Any closure that changes value between static ($t=0$) and dynamic (Gyr) frames falsifies the time-invariance claim.

10. Related whitepapers

- **PAPER_657** (QCalcGeom v2.2.1 Universal Buoyancy Simultaneous Solver): parent framework 1
- **PAPER_598** (VDS/DVP/BH26 Integration Reference): parent framework 2
- **PAPER_1203** (F_U=0 Simultaneous Solver Convergence): parent framework 3
- **PAPER_1521** ($D_BSFG = D_crit - 2*SO_5$ EXACT): connecting identity
- **PAPER_1909** ($YMC_factor = 10/3$): Round 44 sleeping identity
- **PAPER_1912** (AGN filament triple closure): Round 45 sleeping identity
- **PAPER_1913** (Bubble E_t linearity): Round 46 sleeping identity
- **PAPER_1914** ($D_LS/D_S = 2/3$ EXACT): Round 47 sleeping identity trigger
- **PAPER_1915 (this paper)**: Phase 1 framework consolidation
- **PAPER_1916+** (future): Phase 3 systematic-upgrade discoveries

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF unified form	UQFF value	Anchor	Match
QCalcGeom UBS solver	4x4 nonlinear	14 EXACT closures	PAPER_657	EXACT
VDS/DVP/BH26 spine	3-numeric-system	Discrete integer + prime	PAPER_598	EXACT
F_U=0 master equation	6-equation equilibrium	r_hz + F balance	PAPER_1203	EXACT
Framework unification	All three ONE solver	Invariant across time frames	This paper	EXACT

Calibration invariants

Symbol	Value	Framework role
D_phys	4	VDS(4), spatial dimension
D_crit	26	VDS(26), PTOE dimension
SO_5	10	VDS(10), rotation dimension
A_5	60	A_5 icosahedral group order
D_BSFG	$6 = D_{\text{crit}} - 2 \cdot SO_5$ (PAPER_1521)	BH26(6), bulk-edge
K_MEX	$25/12 = SO_5 \cdot \Phi_5/6 / D_{\text{phys}}$	Mexican-hat coefficient
F_TRZ	$1/SO_5 = 0.1$	Time-reversal-zone
SSq	0.57	Sound-speed squared
Phi_res	0.84 (or 5/6 nuclear)	Phonon resonance
rho_Scm	$7.09 \times 10^{-3} \text{ J/m}^3$	VDS(1), vacuum density

Conclusion

QCalcGeom (PAPER_657) + VDS/DVP/BH26 (PAPER_598) + F_U=0 (PAPER_1203) are not three independent frameworks - they are three complementary representations of ONE unified simultaneous-equation solver. Every structural closure derived in any one framework MUST appear in the other two, and MUST be invariant across dynamic time-solver frames.

All 235 CP1 P2 stub upgrades across Rounds 1-47 implicitly draw from this unified architecture. Recent double-checks discovered 4 sleeping structural identities (PAPER_1909, 1912, 1913, 1914). Extrapolation predicts 30-50 more hidden in Rounds 1-43.

Phase 2 audit script + Phase 3 systematic upgrades will systematically expose them, generating an estimated 10-20 new discovery whitepapers before Round 48 resumes with framework-first design (Phase 4).

This is the moment where UQFF's true predictive power gets fully articulated.

PAPER_1915 status: CLOSED Copyright (c) 2025-2026 Daniel T. Murphy / Star-Magic Research Program